

University American College Skopje
School of Computer Science and Information Technology

Academic 2025/26

Version: 15.01.2025

Course Code: Databases (CSCI 2511)
Semester: IV semester, 2025/2026
Year: Spring, 2026
Prerequisites: CSCI 2331
Lecturer / Mentor: Venera Gudachi, PhD

Assistant / Collaborators: Zoran Gichevski, MSc

Duration of Course: 60 hours

Credit Hours: 4 ACTS / 6 ECTS

Schedule hours:

Group 1

Thursday 15:30- 17:30 – lectures / 3A / 3rd

Thursday, 17:30-19:30 – exercises / L2 / 2nd

Group 2

Thursday 15:30- 17:30 – lectures / 3A / 3rd

Tuesday 17:00-19:00 – exercises / L5 / 4th

Consultations hours: Every last Tuesday/Thursday of the month from 17:15 hrs.

Instructor's contact details:

venera.gudachi@uacs.edu.mk /theory

zoran.gichevski@uacs.edu.mk / practice

Required text:

Database System Concepts, 7th Edition, by Abraham Silberschatz,

Henry F. Korth S. Sudarshan

McGraw Hill Education, 2020

ISBN: 978-0-07-802215-9

The book is available in Moodle (Course book)



Supplementary reading:

A First Course in DATABASE Systems, 3rd edition - J. D. Ullman,

J. Widom.

The book is available in hardcopy in UACS library

Additional literature:

Concepts of Database Management, 9th Edition, by Joy L. Starks, Philip J. Pratt, Mary Z. Last, Cengage, 2019, Library of Congress Control Number: 2017963668, ISBN: 978-1-337-09342-2

Database Systems: The Complete Book, 2/E, by Hector Garcia-Molina, Jeff Ullman and Jennifer Widom, Pearson, 2014, ISBN-10: 1-292-02447-X, ISBN-13: 978-1-292-02447-9

Database Concepts, 9th Edition, by David M. Kroenke, David J. Auer, Scott L. Vandenberg, Robert C. Yoder, Pearson, 2020, ISBN-10: 0-13-518814-8, ISBN-13: 978-0-13-518814-9

Course outline/ description:

This course is designed for software engineers, database engineers, and programmers, providing foundational

knowledge in managing relational database systems. It covers essential concepts of database design, with a focus on query processing and optimization, as well as efficient data retrieval.

Students will learn to structure and manage data effectively within relational databases and perform various database management tasks. Additionally, the course aims to develop the skills required to build and manage different types of database applications.

Key topics include relational database design, query optimization, and transaction processing. Other areas of focus include an introduction to database management systems (DBMS), relational models, SQL data definitions, entity-relationship (ER) modeling, UML and ODL, relational algebra, and advanced SQL concepts. Students will also learn about constraints, triggers, transactions, views, indexes, and functional dependencies. The course will address normalization techniques, dependency inheritance, database repair processes, Codd's 12 rules, and essential aspects of database security and authorization.

Learning objectives:

After the completion of the course, the students should gain the knowledge and understanding of:

- Explain the fundamental concepts of database management systems (DBMS) and the relational model.
- Design and implement efficient relational database structures using entity-relationship (ER) modeling and normalization techniques.
- Write SQL queries to retrieve, manipulate, and manage data, including the use of advanced SQL features such as joins, subqueries, triggers, and stored procedures.
- Analyze and optimize database performance through indexing, query optimization, and execution plans.
- Manage database transactions and implement concurrency control to ensure data consistency and integrity.
- Perform essential database maintenance tasks, including backup, recovery, and data security measures.
- Apply authentication, authorization, and encryption techniques to protect database systems and ensure compliance with data privacy regulations.
- Develop practical database-driven applications and solutions for real-world scenarios.
- Critically evaluate different database design strategies and make informed decisions to improve performance, scalability, and security.
- Collaborate effectively on database-related projects, presenting and documenting solutions clearly and professionally.
- Demonstrate lifelong learning skills by exploring emerging trends and advancements in database technologies.

Course delivery and format:

The course will be conducted on-site, with exceptions permitted upon approval. The course will combine lectures, guided exercises, laboratory practice, and AI-assisted learning activities to support both theoretical understanding and practical skills. It is designed to help students develop expertise in advanced SQL programming, database performance optimization, and modern database architectures.

Lectures. Core concepts will be introduced through instructor-led presentations supported by slides and real-world examples. Topics will include advanced SQL constructs (e.g., triggers, stored procedures, functions, and views), query optimization strategies, transaction management and recovery, indexing methods, concurrency control, distributed databases, and security models. Lecture outlines will be shared to guide students in preparing for discussions and exercises.

Practical exercises. Students will solve progressively complex SQL problems, perform query tuning, design indexing strategies, and implement transactions with concurrency control mechanisms. Exercises will also include normalization of large datasets and design of relational schemas. Students will apply theory using MS SQL Express with SQL Management Studio, MySQL with MySQL Workbench, and MongoDB with Mongoshell. Projects will focus on implementing advanced SQL features, analyzing query execution plans, testing recovery scenarios, and developing database-driven applications. Instructor guidance will be available throughout.

Interactive Learning with Mentimeter. To encourage engagement and immediate feedback, Mentimeter will be used during lectures for live polls, knowledge checks, and short case-based discussions. This tool will help students reflect on complex database topics in real time and foster active participation.

AI-Assisted Teaching. Students will be encouraged to use Large Language Models (LLMs) (e.g., ChatGPT, Gemini, Claude or similar tools) to generate SQL query examples, experiment with optimization strategies, or clarify advanced theoretical concepts. AI-generated study aids (summaries, code snippets, or mini-quizzes) may also be provided to strengthen student engagement with complex topics.

Official Quizzes in Moodle. Periodic weekly online quizzes will assess mastery of advanced SQL and database management topics. These will be automatically graded and contribute to the active participation grade.

Independent Study. Students are expected to review academic literature, technical documentation, and advanced database references. Independent work will include reading research papers on database systems, experimenting

with SQL performance tuning, and preparing case studies for class discussion.

Learning outcomes

By the end of this module, students will achieve the following learning outcomes:

- **Knowledge:** Students will develop a comprehensive understanding of database systems, including core concepts such as relational models, SQL, data normalization, indexing, and transaction management. They will become familiar with key principles of database design, query optimization, and data security, as well as the use of constraints, triggers, views, and functional dependencies.
- **Skills:** Students will acquire practical skills in designing, creating, and managing relational databases. They will learn to write efficient SQL queries for data retrieval and manipulation, optimize database performance, and implement transaction management techniques. Through hands-on tasks and projects, students will apply these skills to solve real-world database problems and develop database-driven applications.
- **Competences:** Students will be able to critically evaluate various database design strategies and make informed decisions to improve database performance, integrity, and security. They will demonstrate competence in creating well-structured databases by integrating theoretical knowledge with practical implementation. Furthermore, students will enhance their communication and teamwork skills by presenting their work in projects, writing detailed reports, and participating in peer reviews. Through independent project work, they will also develop the ability to engage in lifelong learning and prepare for future academic or professional challenges.

Course delivery plan:

Class	Month	Chapter	Lecture Topics	Exercises Topics
1	Week 1	1	Introduction to DBMS's, Relational Model, SQL Data Definition	Introduction
2	Week 2	2	Entity/Relationship Model	Exercises on the subject
3.	Week 3	3	More E/R, UML, ODL	Exercises on the subject
4.	Week 4	4	Relational data model, Relational algebra	Exercises on the subject
5.	Week 5	5	Introduction to SQL	Exercises on the subject
6.	Week 6	6	Advanced Relational Algebra and SQL	Exercises on the subject
7.	Week 7		Mid Term Exam	Mid Term Exam
8.	Week 8		Constraints and Triggers	Exercises on the subject
9.	Week 9	7	SQL Transactions, Views, and Indexes	Exercises on the subject
10	Week 10	8	Functional Dependencies, Normalization	Exercises on the subject
11.	Week 11	9	Inference of Dependencies, Multival Dependencies, 3NF, 4NF	Exercises on the subject
12.	Week 12	10	Reparation of a database. Codd's 12 Rules.	Exercises on the subject
13.	Week 13	11	Authorization, Security	Exercises on the subject
14.	Week 14	12	Project presentation	Exercises on the subject
15.	Week 15	13	Reading / Final Exam Preparation	Reading week/ Solving exam Problems
16	Week 16		Final exam	Final Exam

Instructor's expectations from students/Students' responsibilities:

- Regular attendance of lectures and exercises on-site.
- Timely arrival at classes.
- Careful follow-up of lectures and exercises.
- Academic behavior and preservation of the working atmosphere of classes.
- Mandatory remarks about everything that is written and spoken in classes.
- Active participation in the discussion of the lessons by completing the provided quizzes.
- Timely preparation and mandatory oral presentation of semestral projects if given.

- Independence and honesty in work.
- Avoid anti-social behavior in college and anti-academic behavior in the classroom (i.e., plagiarism, cheating, etc.)
- Impose a self – discipline regarding college’s rules and procedures.
- If online sessions are scheduled on special occasions, the students are required to turn On their cameras at all times during the class.

AI Statement on Generative AI phenomenon

During class sessions, generative AI tools may be used to support learning, providing an opportunity to explore their benefits and limitations. As learning to use AI is an emerging skill, the implications of these systems will be examined together in class. However, AI use will be limited to specific in-class exercises, such as brainstorming ideas, developing outlines, or generating ways to discuss a problem. The instructor will clearly indicate when and where AI use is appropriate. It is important to note that AI tools may not be used to generate work for graded assignments, as they do not substitute for the development of fundamental skills aligned with the course’s learning objectives. AI-generated content may be incomplete, inaccurate, or biased and should not be relied upon as a primary method for research.

For assignments, the use of AI tools such as ChatGPT is prohibited unless explicitly permitted by the instructor in specific contexts, such as coding exercises or research papers. Unauthorized use of AI tools will be treated as plagiarism and subject to appropriate penalties. Students are responsible for the content they submit and may not pass off AI-generated work as their own.

Evaluation structure and assessments during the teaching semester:

Categories in the assessment:

1. *Attendance - 10%*
2. *Activity/Participation - 10%*
3. *Assignment/ project- 20%*
4. *Midterm Exam – 30 % and*
5. *Final Exam 30%*

Examinations will take place on-site without exceptions. All work should be performed in lab computers (no personal laptops are allowed). Examinations in labs will be performed in Safe Exam Browser.

1. **Attendance (10%):** Up to 25% absences will still allow students to earn points for attendance. For absences between 25-50%, no attendance points will be awarded, but students will still be permitted to take the Final Exam. If a student exceeds 50% absences, they will not be allowed to take the Final Exam. Absences must be approved by both the lecturer and the administration, with mutual notification.
2. **Activity/Participation (10%):** Engaging in class discussions, group activities, and practical exercises will positively impact the participation grade. Contributions through interactive tools like Mentimeter will also be considered part of a student's participation. Continuous and meaningful involvement in class activities will be rewarded as part of the grading criteria.
3. **Assignment/ project (20%):** The assignments and projects will be focusing on applying the concepts learned throughout the course.
4. **Midterm Exam (30%):** Will be covering both theoretical concepts (multiple-choice, essay, and true/false questions) and practical problems related to the course material.
5. **Final Exam (30%):** Will be including questions on theory (multiple-choice, essay, and true/false) as well as practical problems requiring the application of the tools and techniques taught in the course.

Grading Criteria		
Grading	% Qty	Points
A-	96-100	4.00
A-	90-95	3.67
B	83-86	3.00
B-	80-82	2.67
C+	77-79	2.33
C	73-76	2.00
C-	70-72	1.67
D+	67-69	1.33
D	63-66	1.00
D-	60-62	0.67
F	0 -59	0.00

IMPORTANT:

- The summative grades and minimum required performances are a minimum 20% of the Midterm Exam and Final Exam points.
- Each category in the assessment is equally distributed between theoretical and practical part, i.e., each 50%. To be able to pass the exam in each category in the assessment there is a need to obtain at least 20% both in theoretical and practical part.

Evaluation structure and assessments for Make-Up examinations

Examinations will take place on-site without exceptions. All work should be performed in lab computers (no

personal laptops are allowed). Examinations in labs will be performed in Safe Exam Browser.

Categories in the assessment:

- *Theoretical part -40%*
- *Practical part -40%*
- *Assignment/ project- 20%*

No other categories will be taken into consideration such as attendance and activity/participation

3) Part-time students

The exam will include questions on the theory taught (multiple -choice, T/F and Critical response Essay). Whereas in the practical part will include exercises from the material.

Categories in the assessment:

- *Theoretical part 50%*
- *Practical part 50%*

Exams are closed books. The professor has the right to contact the student to take an extra-oral exam if needed to determine the final grade.

Mobile phones are strictly not tolerated in the class for any use (including computations). Active participation is meant as the effort and the interest that a student shows in the class, including homework.

After each session students are expected to study all the relevant materials, read all the associated exercises, identify the difficult points and pose their questions in the next session either directly to me or in the class.

Cheating and plagiarism in any form will result immediately in the grade F. The grade of (incomplete) is assigned only when a student misses the Midterm or the Final exam due to a **college-excused absence**, and is unable to make up before final grades, and are due to the Record's Office. Please, consider seriously that there will be no exceptions concerning the above policy.

Class Conduct

Students are held responsible for everything that is announced, presented or discussed in class. The way to avoid any misunderstanding associated with this course is to attend classes. Students are expected to attend class regularly since attendance records will be kept. Late arrivals create distraction resulting in decline in the quality of the lectures.

Importantly, coming lates to class is also disruptive to other student's colleagues and will be tolerated only up to five minutes late at the beginning of the class. Please, refrain from talking during class, it is disruptive to everyone. The best behavior from all students is expected as that what education is all about. In case there is a question about the material, please don't hesitate to ask the instructor and/or the assistant instructor; answering questions is part of our job. In case someone feels lost or confused, making questions may help clarify the topic. Please, consider that the language of instruction is English, so all our conversation into the class must be in this language.

Keep in mind the following

C- or better is required to use a course either as a prerequisite or as a major requirement.

G.P.A. (Grade Point Average): is computed for each student using the quality points earned for each course taken.

A G.P.A of at least 1.67 is required for transfer and to graduate from ACS.

Academic Honesty

Academic honesty is a core value that safeguards the reputation of University American College Skopje and its faculty, staff, and students.

All UACS members are expected to conduct themselves in an honest manner observing the University Code of Ethics. Plagiarism, cheating, multiple submissions, falsification of data, and unauthorized assistance are considered as serious offences and may result in grade F (fail) on the exam, project, and course, and/or lead to further sanctions, including expulsion from the University. Any work that is submitted at any stage must be your own. Any words, ideas, or data borrowed from other sources must be properly credited and documented.

You may not use generative AI tools, such as ChatGPT, while completing any assignment, unless explicitly permitted by the course instructor in specific contexts. Any use of generative AI tools outside of these parameters constitutes plagiarism and will be sanctioned accordingly.

Late Work

Many reasons are given for late work (e.g., flat tires, printer problems, illness, oversleeping, etc.). No matter what the reason, the bottom line is that either (a) the work was completed and turned in when required or (b) the

assignment was not completed and turned in when required.

As specified earlier, assignments are due in class at the beginning of class on the specified due date. There is NO PROVISION for late work on any assignment (i.e., late work is not accepted). It is strongly recommended to have backup systems in place so that students can have all work completed on schedule. Having the work completed on schedule is a key to early success in student's business career.

NOTE: Late submissions are not accepted. Partial credit will NOT be given for late work.

Wishing you an interesting and creative academic semester.

Prepared by: Venera Gudachi, Ph.D. and Zoran Gichevski, MSc